each open item is stated in Section 16 with its nearest proved theorem and missing obligation

established (Papers I--III; SIV-P1 to P9)

- endpoint and occurrence projections factor (P1, P2)
- realised set monotone; primitive status not (P3)
- swap classes: arithmetic, non-monotone (P4)
- endpoint geometry misses ancestry (P5)
- persistence hierarchy $D \supseteq$ components $\supseteq$ edges (P6)
- local inventory insufficient for $\Delta\beta$ (P7)
- scaling data insufficient for β (P8)
- levels L1-L5 assembled (P9)

open construction obligations

O1 classify history complexes in general

O2 ancestry from richer invariants

O3 maps history geometry $\leftrightarrow$ occurrence geometry

O4 predict $\Delta\beta_{\geq 2}$ without full target

O5 onset bounds and Betti laws for towers

O6 reachability of activation families

O7 transitions of dormant primitive systems

O8 compatibility of arithmetic guards